

Students will calculate the perimeter & area of polygons.

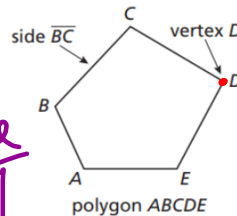
Do Now

In the space below, list all the information that you know about the following shapes:

- Triangle
 - 3 sides and corners
 - 3 $\angle$ s add to 180°
 - 3 types: equilateral, isosceles, acute
- Rectangle
 - 2 pairs of $\cong$ sides
 - 4 sides
 - 4 rt. $\angle$ s
- Square
 - 4 $\cong$ sides
 - 4 rt. $\angle$ s
 - rectangle
- Parallelogram
 - 4 sides
 - opposite sides are $\parallel$
- Polygon
 - shape w/ sides + angles
 - 3D?

Polygons

A many **poly**gon is a figure closed by three or more sides (denoted by segments). Each side intersects exactly two other sides at a common endpoint called a vertex.



↑ Vertices (many)

Decade

Below are the names of common polygons we will use in this course.

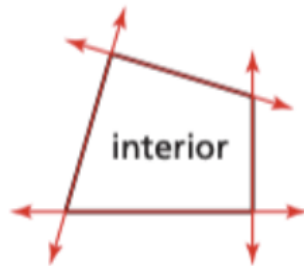
# of Sides	Name	# of Sides	Name
3	Triangle	8	Octagon
4	Quadrilateral	9	Nonagon
5	Pentagon	10	Decagon
6	Hexagon	12	Dodecagon
7	Heptagon	n	n -agon

(Septagon)

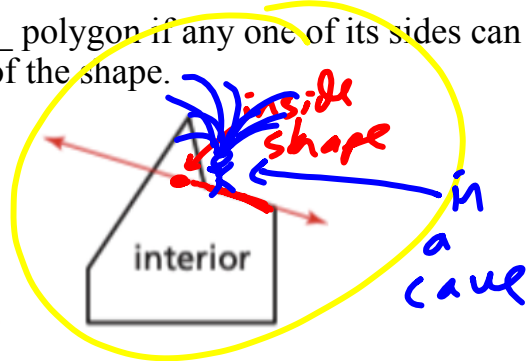
11-gon

Convex vs Concave Polygons

A polygon is called a Concave polygon if any one of its sides can be extended so that it enters inside of the shape.



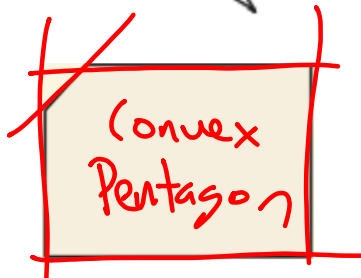
Convex



Concave

Polygons that are not concave are called convex polygons. In these polygons, it is impossible to extend a side so that it enters the inside of the shape.

Ex 1: Classify each polygon by its number of sides and determine whether it is convex or concave.



Finding Perimeter & Area of Polygons

The **perimeter** of any polygon is the distance around the shape. This tells us how far (in *meters*, *miles*, *feet*, etc.) is needed to walk around the shape.

The **area** measures the space inside the polygon. This measure will always be given in units squared (m^2 , mi^2 , ft^2 , etc.).

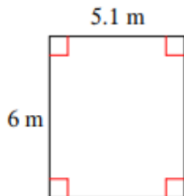
Here is how to calculate the perimeter and area of common polygons:

Name	Triangle	Square	Rectangle	Parallelogram
Diagram				
Perimeter	$P = c + a + b$	$P = s + s + s + s$	$P = w + l + w + l$	
Area	$A = \frac{1}{2} \cdot b \cdot h$	$A = b \cdot h$	$A = b \cdot h$	$A = b \cdot h$

must make a 90° $\angle$.

Find the perimeter and area of each polygon below.

1)



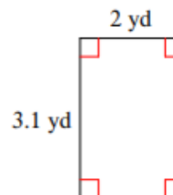
Perimeter:

$$22.2 \text{ m}$$

Area:

$$30.6 \text{ m}^2$$

2)



Perimeter:

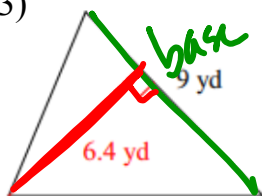
$$10.2 \text{ yd}$$

Area:

$$6.2 \text{ yd}^2$$

Find the area of each polygon below.

3)



$$\begin{aligned}
 A &= \frac{1}{2} b \cdot h \\
 &= \frac{1}{2} \cdot 9 \cdot 6.4 \\
 &= 28.8 \text{ yd}^2
 \end{aligned}$$

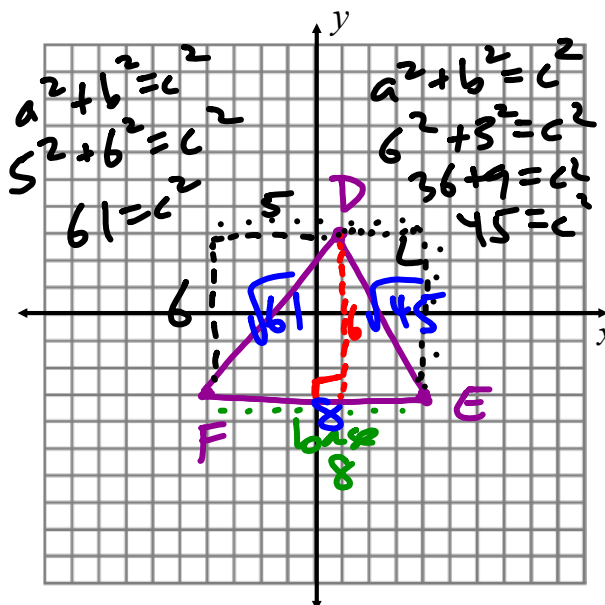
4)



$$\begin{aligned}
 A &= b \cdot h \\
 &= 8 \cdot 5.7 \\
 &= 45.6 \text{ m}^2
 \end{aligned}$$

Perimeter & Area in the Coordinate Plane

Ex 2: Find the perimeter and area of $\triangle DEF$ with coordinates $D(1,3)$, $E(4,-3)$, and $F(-4,-3)$.

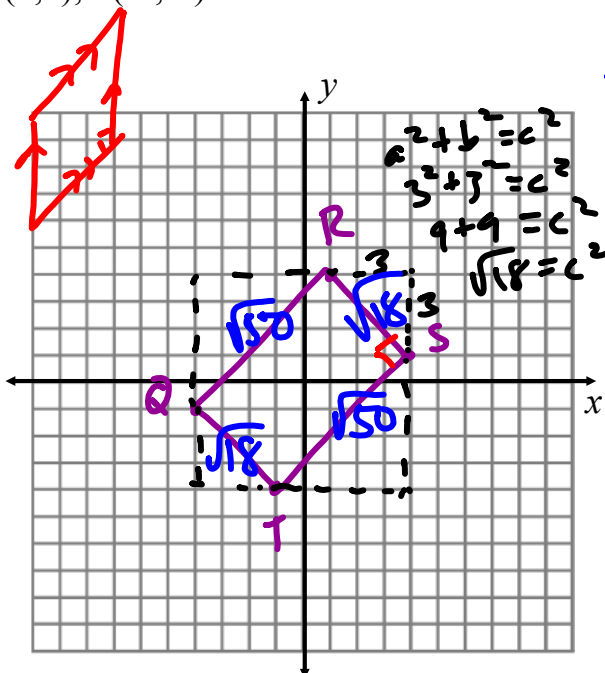


$$\begin{aligned}
 P &= 8 + \sqrt{45} + \sqrt{61} \\
 &\approx 22.5 \text{ units}
 \end{aligned}$$

$$\begin{aligned}
 A &= \frac{1}{2} b \cdot h \\
 &= \frac{1}{2} \cdot 8 \cdot 6 \\
 &= 24 \text{ units}^2
 \end{aligned}$$

$$8 + \sqrt{45} + \sqrt{65}$$

5) Find the perimeter and area of $QRST$ with coordinates $Q(-4,-1)$, $R(1,4)$, $S(4,1)$, $T(-1,-4)$.



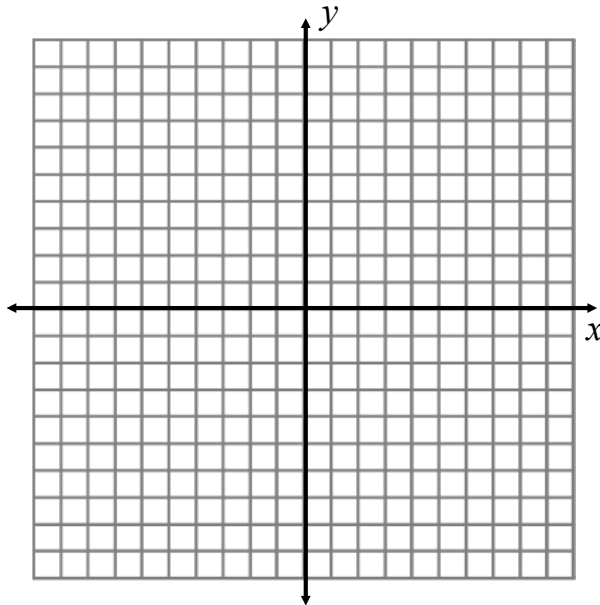
P:

$$P = \sqrt{18} + \sqrt{50} + \sqrt{18} + \sqrt{50}$$

A:

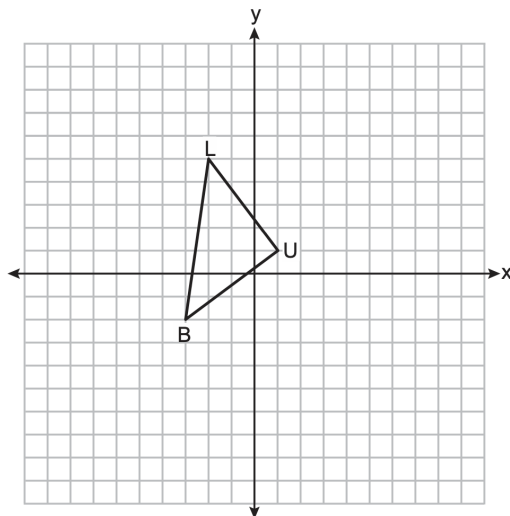
$$A = \sqrt{18} \sqrt{50} = 30$$

- 6) Find the perimeter and area of $JKLM$ with coordinates $J(-3,5)$, $K(1,5)$, $S(2,-1)$, $M(-2,-1)$.



Challenge

- 7) On the set of axes below, $\triangle BLU$ has vertices with coordinates $B(-3,-2)$, $L(-2,5)$, and $U(1,1)$.



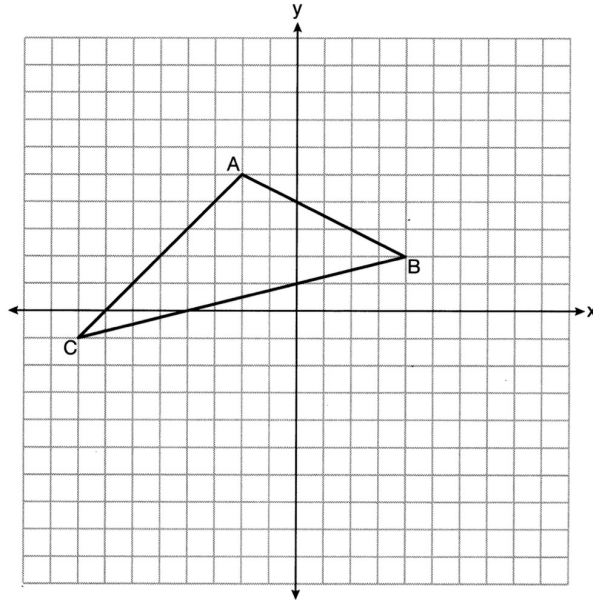
What is the area of $\triangle BLU$?

- | | |
|----------|----------|
| (1) 11 | (3) 14 |
| (2) 12.5 | (4) 17.1 |

8)

Challenge

Triangle ABC with coordinates $A(-2,5)$, $B(4,2)$, and $C(-8,-1)$ is graphed on the set of axes below.



Determine and state the area of $\triangle ABC$.

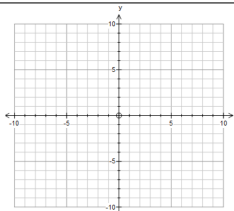
Geometry

Name: _____ Hr: _____

HW 1.5 Perimeter and Area

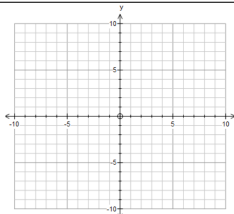
- 1) Find the perimeter and area of quadrilateral $PQRS$ with the vertices $P(2, 4)$, $Q(2, 3)$, $R(-2, -2)$, and $S(-2, 3)$.

Perimeter= _____ Area = _____



- 2) Find the perimeter and area of the polygon with the given vertices: $T(0, -2)$, $U(3, 5)$, $V(-3, 5)$

Perimeter= _____ Area = _____

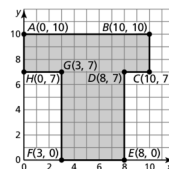


- 3) You are buying tile for your bathroom floor and baseboards for your bathroom walls. In the figure, the entire polygon represents the layout of the floor. Each unit in the coordinate plane represents 1 foot.

a) Find the area of the floor.

b) Find the perimeter of the floor.

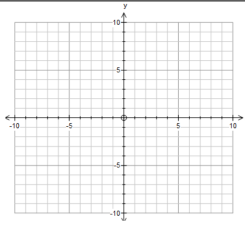
c) The cost of the baseboard is \$2 per foot. The cost of the tile is \$2.50 per square foot. Find the total cost to buy tile and baseboards for your bathroom.



- 4) Find the perimeter and area of the polygon with the give vertices:

$A(3, 7), B(5, 7), C(3, -7), D(5, -7)$

Perimeter= _____ Area = _____



- 5) Find the perimeter of the square ADEF.

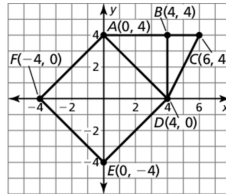
- 6) Find the perimeter of the triangle BCD.

- 7) Find the area of square ADEF.

- 8) Find the area of triangle ACD.

- 9) Find the area of triangle BCD.

- 10) Find the area of pentagon ACDEF.



- 11) A rectangle has vertices $(1, 4), (3, 4),$ and $(3, -3)$. Find the remaining vertex of the rectangle. What is the perimeter and area of the rectangle?

Perimeter= _____ Area = _____

